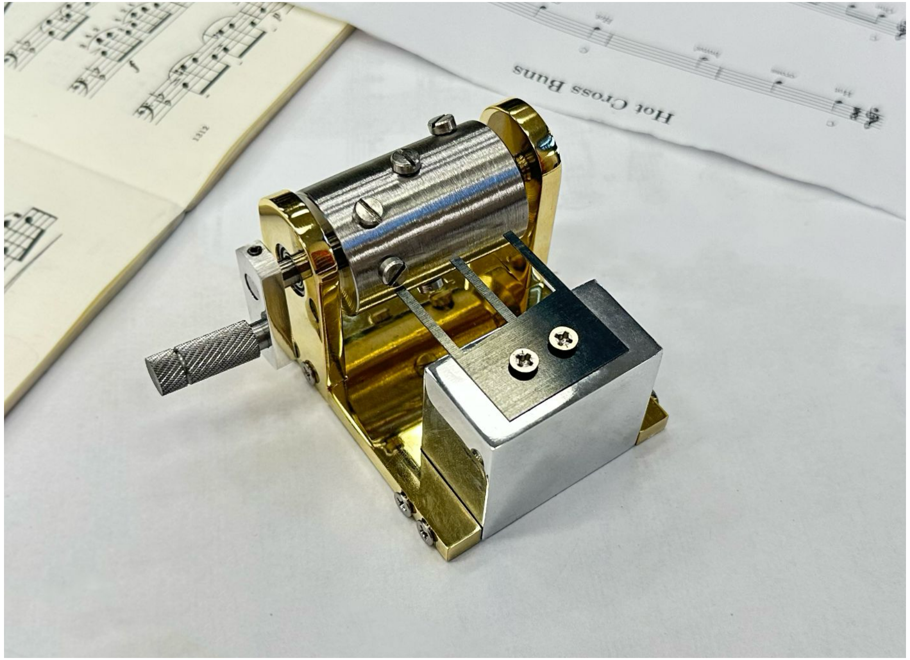
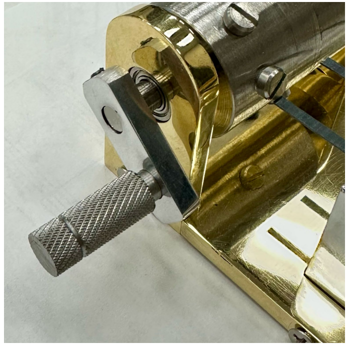
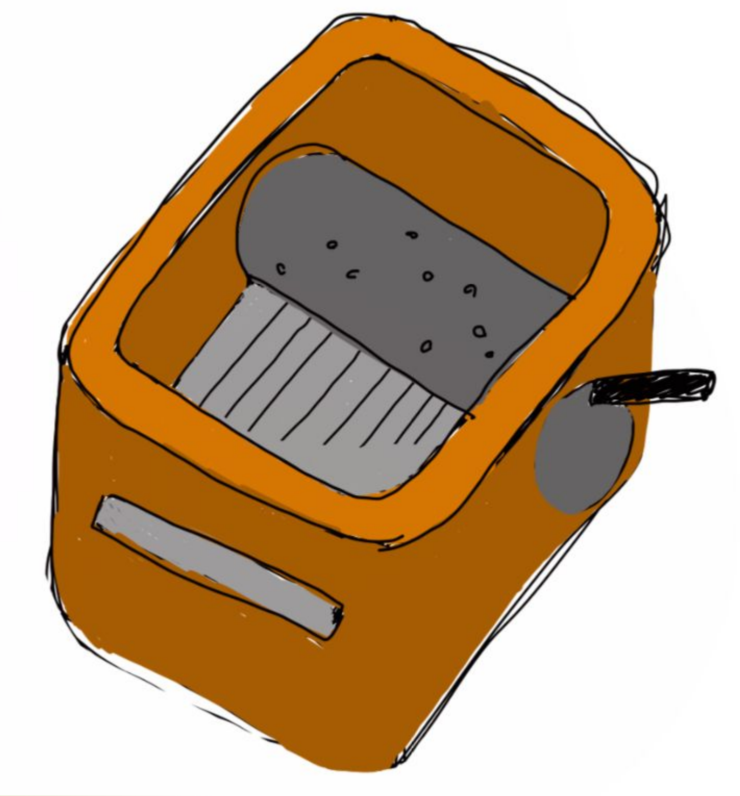
The (Modern) Music Box

Madison Lee | ME103 | Winter 2026



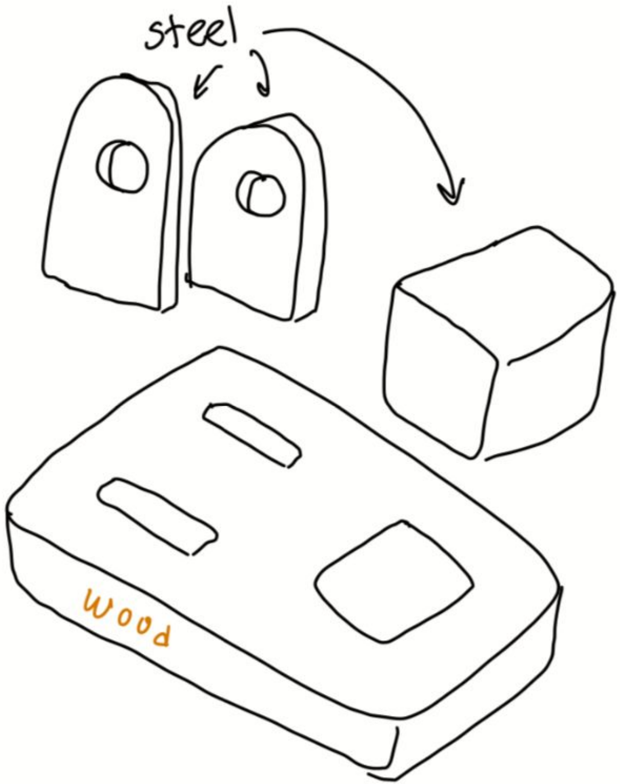


Ideation – Sketch 1

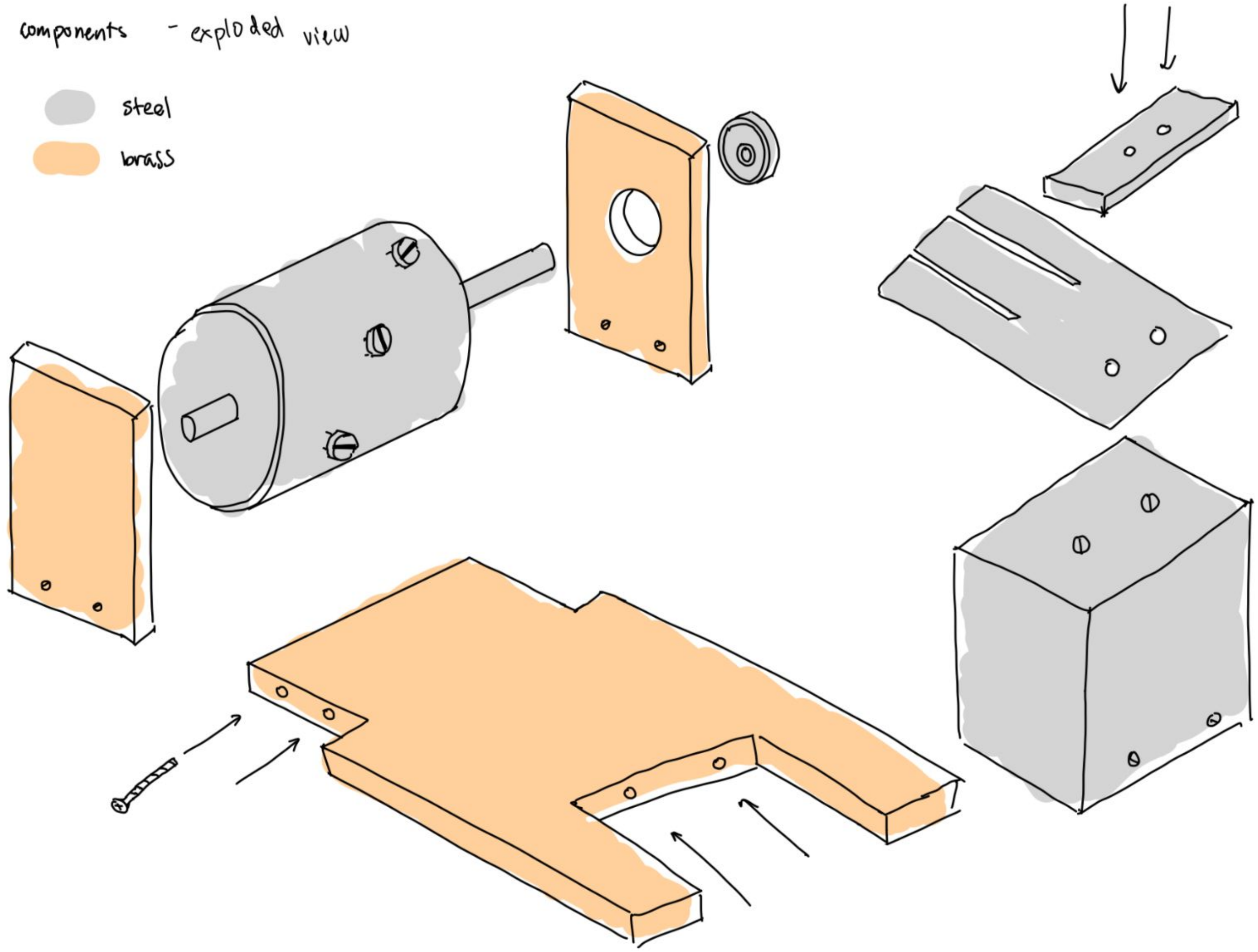


- A lot of interactions
  - Each component was heavily dependent on other components
- Too many parts & materials

Ideation – Sketch 2



- Scoped down from initial sketch
- Uncertainty on connections/fasteners & materials

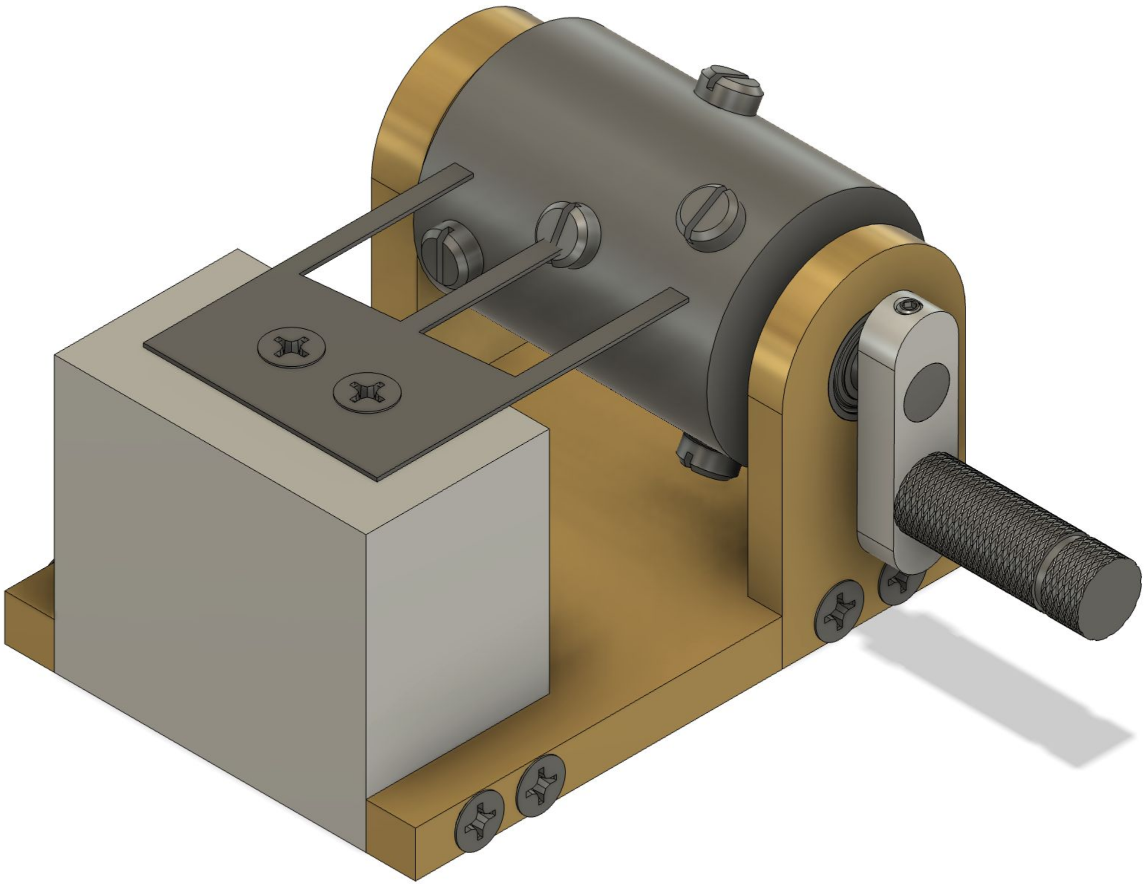


components - exploded view
steel
brass

Final CAD

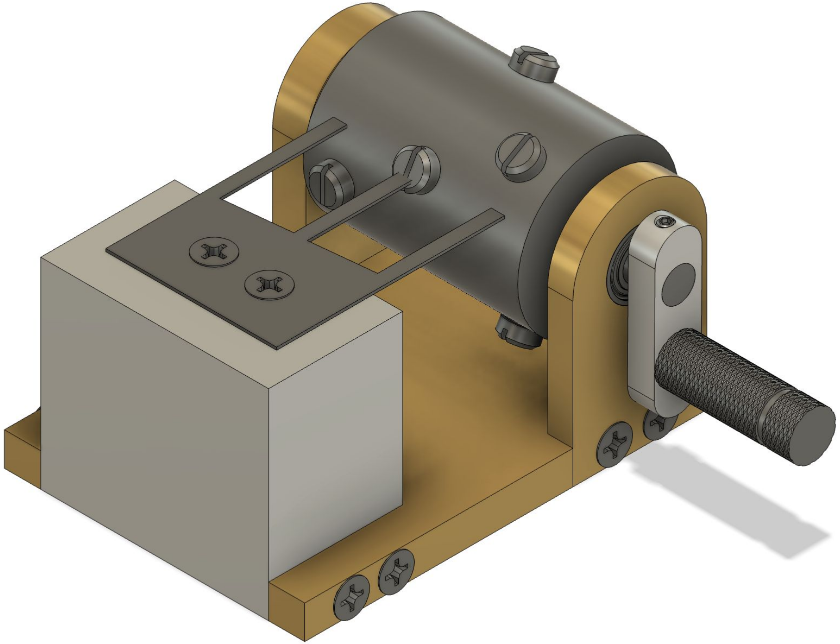
Materials: Aluminum, Steel, Brass

Fasteners: 6-32 0.5” screws, 6-32 set screws, Cheese-head slotted screws, 0.5” OD ball bearing



Operation Sequence

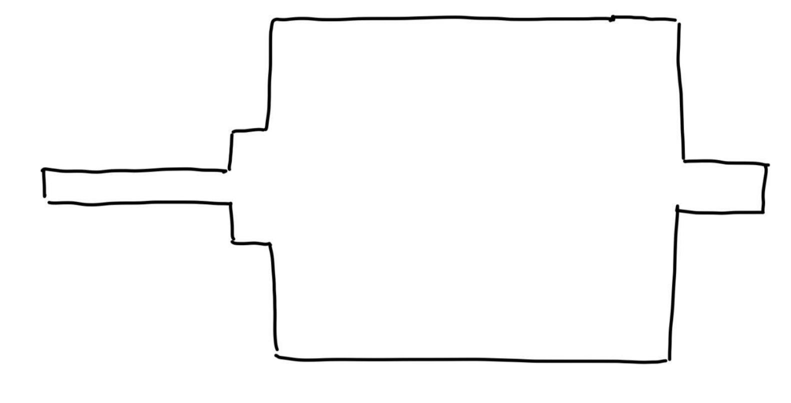
1	Barrel	Lathe
2	Legs	Mill
3	Tines Base	Mill
4	Base	Mill
5	Handle	Lathe
6	Handle Connection	Mill
7	Tines	Laser cutter

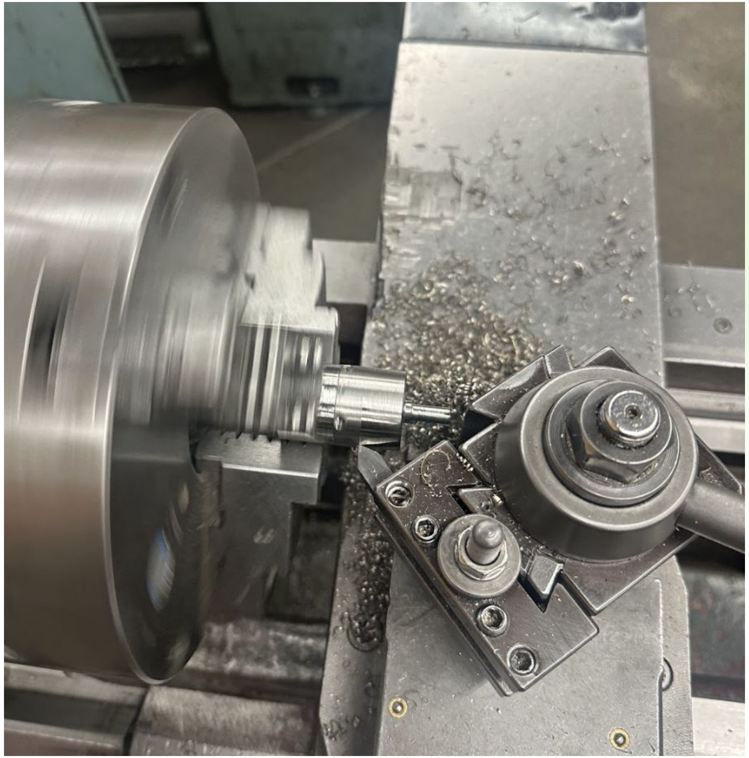


Critical Component #1: Barrel

Lathe

- Tolerance was ~0 because the rods had to friction fit into a ball bearing
- Machine finish: Had to cut the way I wanted to present it

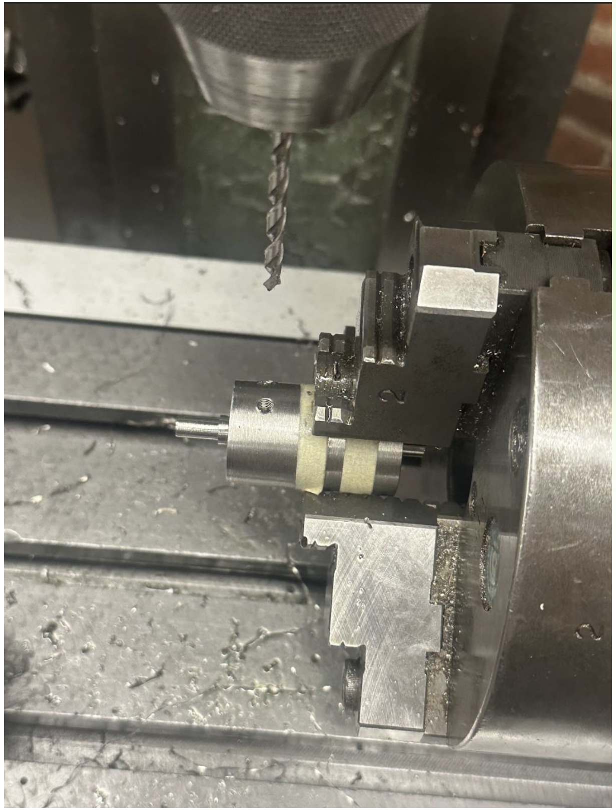




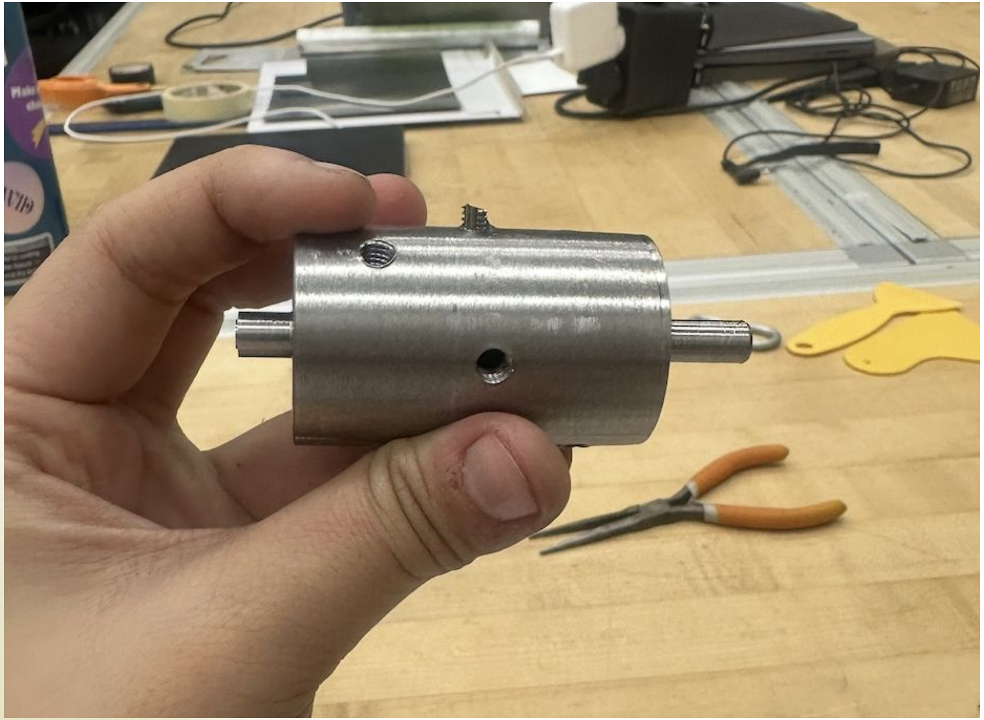
Critical Component #1: Barrel

Mill – Indexing head

- Cutting through steel: SUPER SLOW PROCESS
- Difficult to not rotate when moving onto the next key



Critical Component #1: Barrel



How are actual tines made?

- Cut from a single piece of high-carbon steel
  - CNC Machines to cut the width & thicknesses of each key
- Hardened & Tempered
  - To ensure clear vibration, steel undergoes heat treatment, and then is quenched to harden it
- Weighted Tuning
  - Solder small amounts of lead or tin to the undersides of the longer teeth to help with consistent vibration

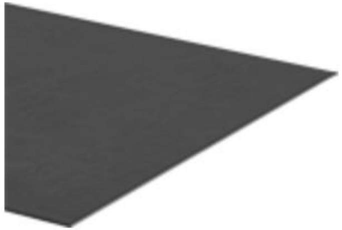
Calculating the length of the tines

$$f_n = \frac{k_n^2}{2\pi L^2}\sqrt{\frac{EI}{\mu}}$$

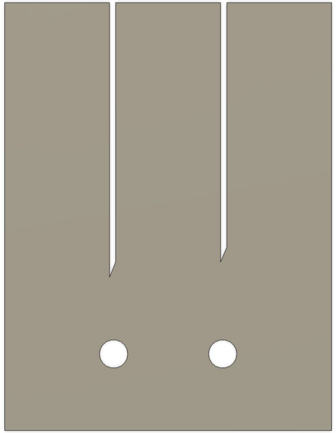
LENGTHS (in)	LENGTHS (cm)	NOTES	FREQUENCY
1.899453957	4.824610445	C4	261.63
1.792875547	4.553901431	D4	293.66
1.692229166	4.298259761	E4	329.63
1.644056616	4.175901551	F4	349.23
1.551777704	3.941513241	G4	392
1.464691641	3.720314759	A4	440
1.382489436	3.511521272	B4	493.88
1.343129608	3.411547362	C5	523.25
1.267743665	3.22006717	D5	587.33
1.196586719	3.039328625	E5	659.26
1.162523582	2.952808304	F5	698.46
1.097279536	2.787088516	G5	783.99
1.035693392	2.630659794	A5	880
0.977562707	2.483007935	B5	987.77
0.9497360537	2.412328274	C6	1046.5
0.8964301422	2.276931332	D6	1174.66
0.8461177918	2.149138031	E6	1318.51
0.8220312505	2.087958249	F6	1396.91
0.7758938006	1.970769189	G6	1567.98
0.7323458205	1.860157379	A6	1760
0.6912429686	1.755756192	B6	1975.53
0.6715648039	1.705773681	C7	2093
0.6338718324	1.610033585	D7	2349.32
0.5982956282	1.519670075	E7	2637.02
0.5812628313	1.476406794	F7	2793.83
0.5486397679	1.393544258	G7	3135.96
0.5178466958	1.315329897	A7	3520
0.488781972	1.241505539	B7	3951.07
0.4748674596	1.206162696	C8	4186.01
0.4482150711	1.138465666	D8	4698.64
0.4234769554	1.075630886	E8	5263.632

Trial 1:

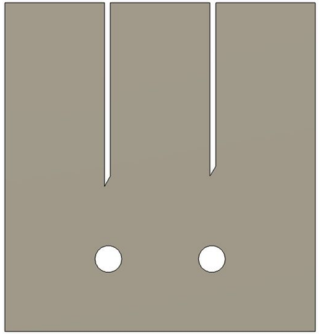
Easy-to-Form 1075 Spring Steel Sheets



- Yield Strength: 50,000 psi
- Hardness: Rockwell B90 (Medium)
- Heat Treatable: Yes
- Max. Hardness After Heat Treatment: Rockwell C64
- Specifications Met: ASTM A684



CDE5,6 @ uniform design



Challenges:

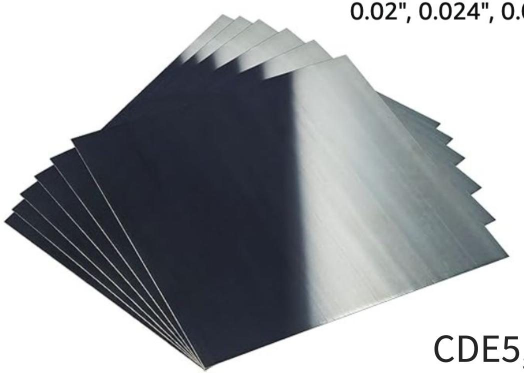
- Required a LOT of force to “pluck” the tines & to hold the tines down
- Produced a very dull sound → “dwang”-y

Positives:

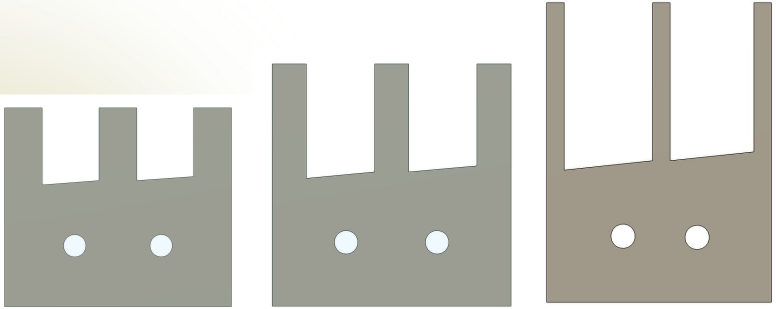
- Didn’t need to be precise on design

Trial 2:

Spring Steel 1095 Shim Sheet Assortment, Blue Tempered, Polished Finish, Spring Steel Shim Stock, 0.004", 0.008", 0.015", 0.02", 0.024", 0.03" Thick, 6" Width, 6" Length, 6 Pieces



CDE5, 6, 7, 8 @ Varying thicknesses



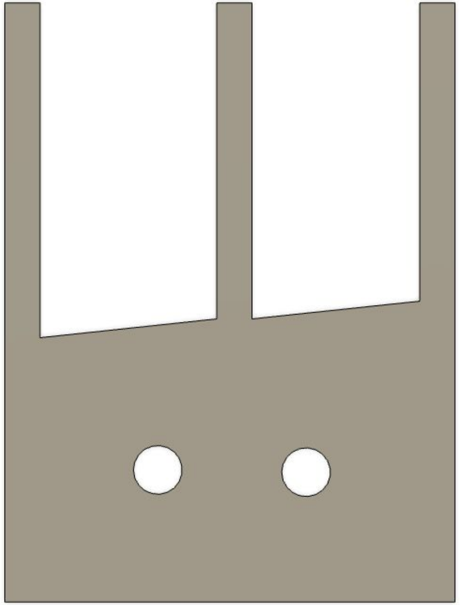
Challenges:

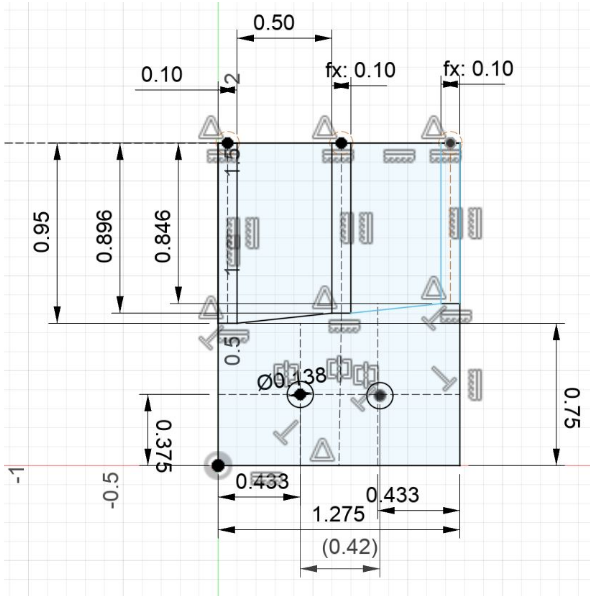
- Design had to very accurately line up with the tines
- Still produced a “twang”-y sound
- Still required a lot of force to hold down the tines to produce a clear sound

Positives:

- Much easier to pluck and hear noise
- Didn’t have to heat treat the steel

Final tine design

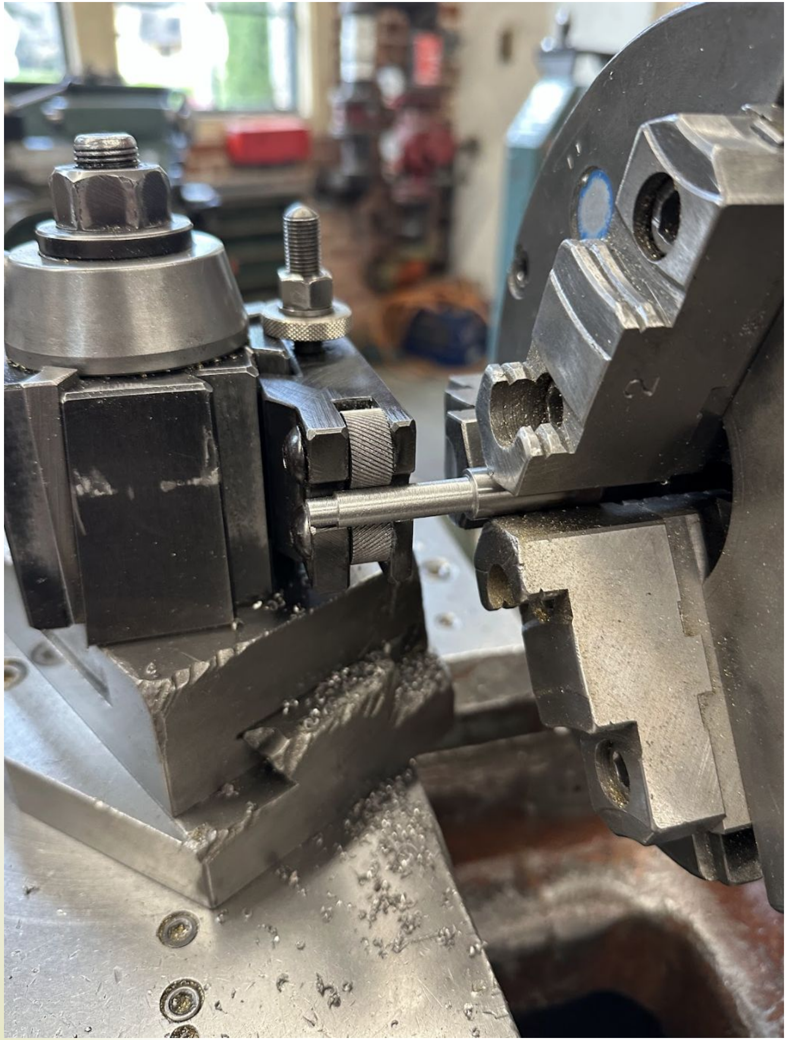


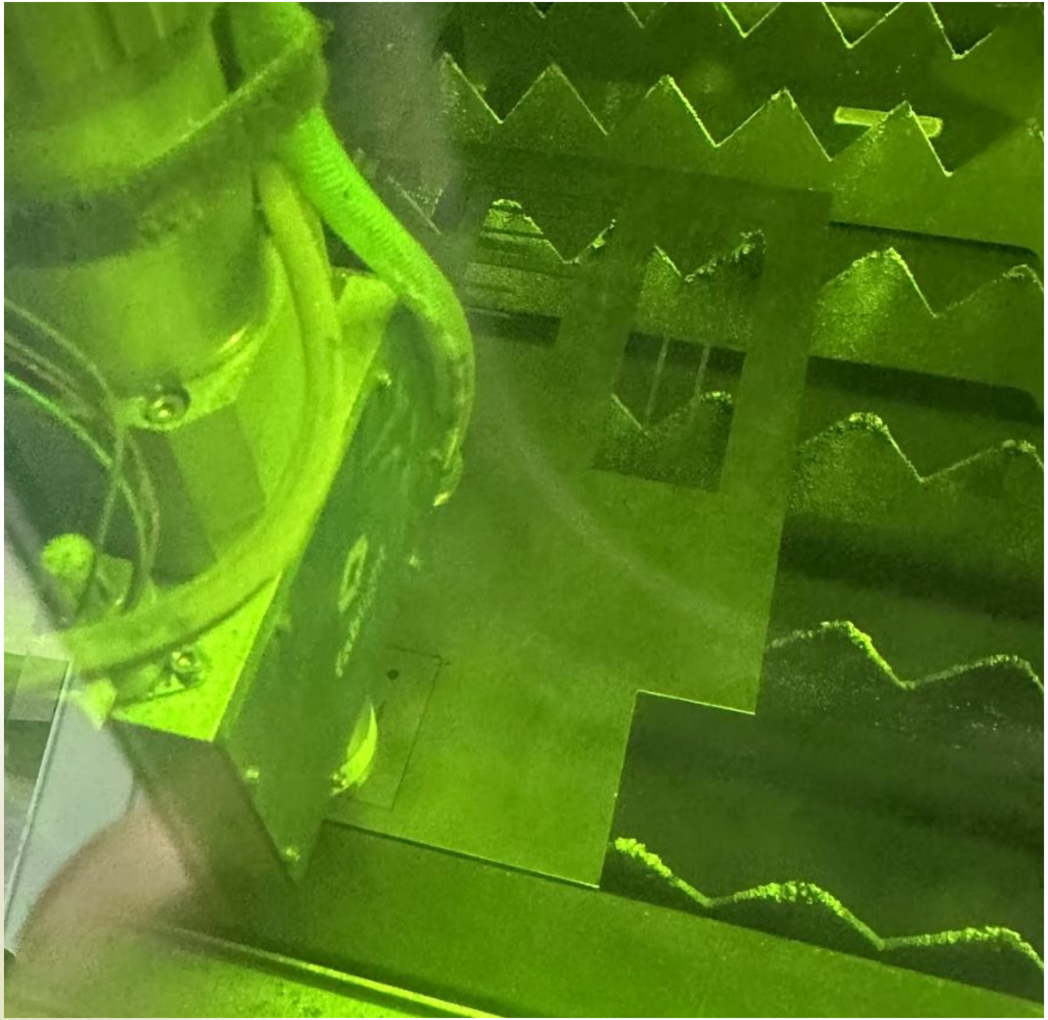


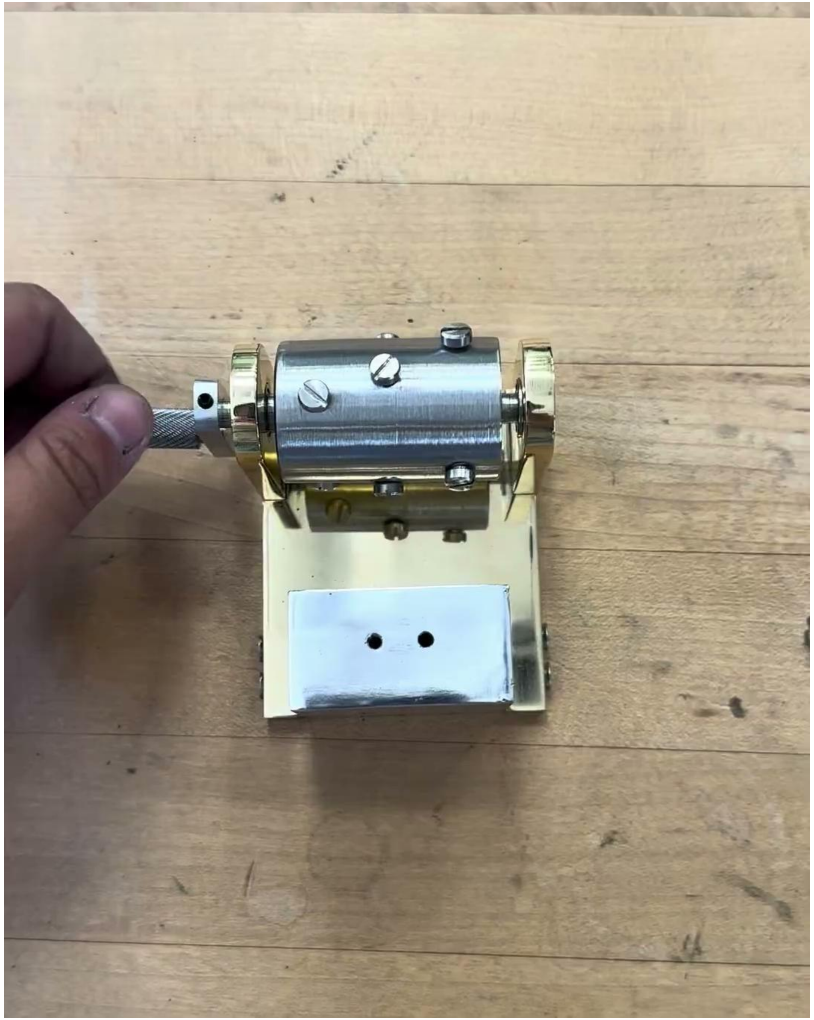
CDE octave 6 using 0.02" blue tempered steel

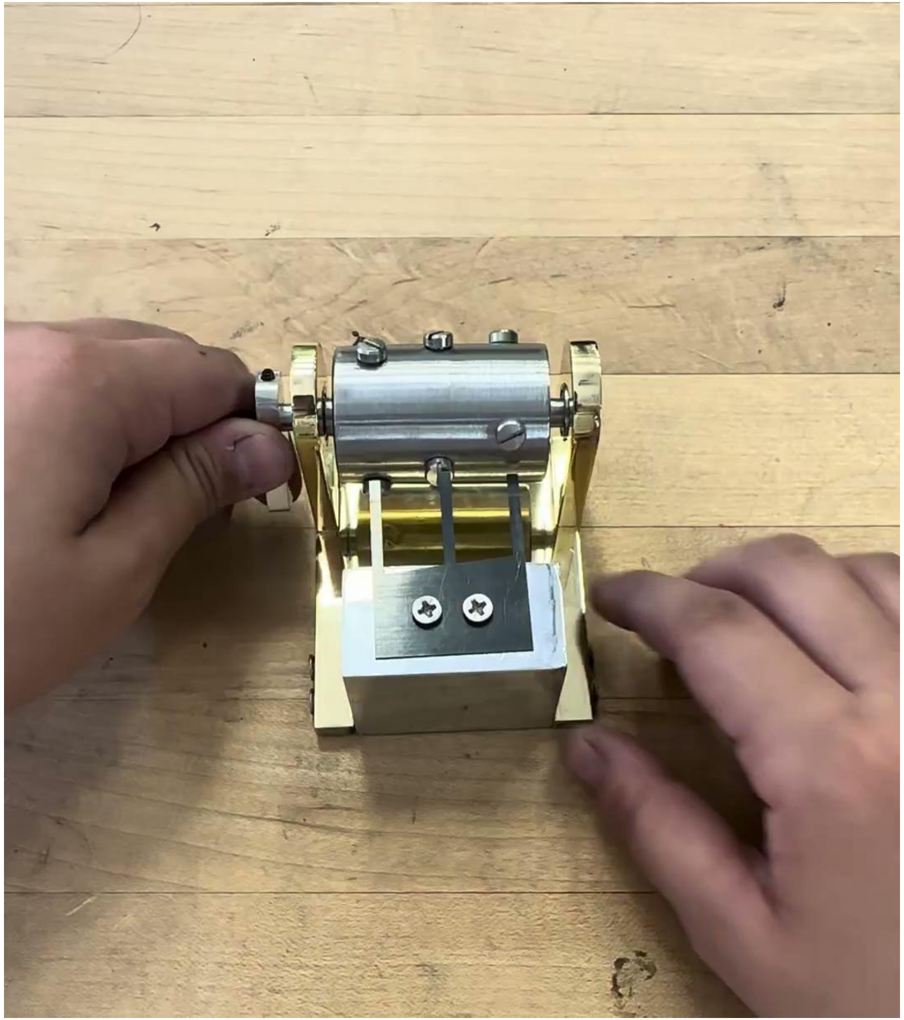

WALL ANCHORS


KURT







Being ready to pivot

From the design, materials, fasteners, throughout the stages of ideation to finalization, nearly every single part of my project changed. I realized how important it was to start with a robust design that was able to quickly and easily adapt to change

Planning

With so many parts that were dependent on each other, I HAD to plan if I wanted to finish on time. I *very* quickly learned that using the mill and the lathe used up more time than initially planned, which made me come up with a realistic plan and timeline.

Just go for it – but prototype first

As important as it was to plan, there were so many moments where I just had to make a decision and go with it. Even if the parts weren't the most perfect, I had to decide on it in order to make the other parts. That being said, I learned the importance of working with scraps. If I had scraps lying around, before making the finalized part, I would test out the steps I would have to go through to see what parts I had to be careful about.

Thank you!